EPSR-test-ex119 Coordination-number constraints: over-bonded secondary atoms

Program(s): epsr | chained passes: 3 | verification: ALL IDENTICAL (all 3 passes)

What this case tests

Exercises the bond-balancing passes of the coordination constraints of the `epsr` engine [1,2,3] in the case where far fewer partner atoms than constrained atoms exist, so partners are heavily over-subscribed.

Code exercised

Fortran (original, fortran_src/):

- `coordconstraint_mod.f90`: `init_constraint`, `find_constraints`, `en_coordcon`, `en_mole_coord`, `check_coord_bonded`, `save_coord_energies`, `writecoordcons`
- `molfit.f`: `en_harmonic`, `en_gauss`, `en_atom_inter_scale_coordextra`, `en_change`
- `ato_manipulation_rev4.f`: `fmolec1`, `press`; `read_and_save_files.f`: `read_pcof`, `save_files_others`

Rust (port, rust/epsr-core/src/):

- `coordconstraint.rs`: `init_constraint`, `find_constraints`, `en_coordcon`, `en_mole_coord`, `check_coord_bonded`, `save_coord_energies`, `writecoordcons`, `write_pcof_coordcons`
- `coordconstraint.rs`: `en_harmonic`, `en_gauss`, `en_atom_inter_scale_coordextra`
- `energy.rs`: `fmolec1`, `en_change`, `press`; `pcof.rs`: `read_pcof`

Physical system

The EPSR-test system: a liquid mixture of 200 dimethyl sulphoxide (DMSO, (CH3)2SO) and 400 water molecules (DMSO mole fraction 1/3; 3200 atoms) at 300 K in a cubic box of side 33.12 Å (atomic number density 0.0881 atoms/Å³). DMSO uses EPSR's Lennard-Jones + Coulomb reference potential with methyl rotational groups; water uses SPC/E Lennard-Jones and charge parameters [4] with EPSR's flexible intramolecular geometry (O-H 0.976 Å). The empirical potential is refined against three neutron diffraction data sets of deuterated DMSO in D2O, in null-scattering H2O/D2O and in H2O (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), the isotope-substitution approach of Soper and Luzar [5]. A `coordcon atc nc cctemp ccdistort ccgaussweight ccgausswidth ats ds dsmax ...` line asks each atom of type `atc` to have `nc` neighbours of type `ats` on other molecules; `find_constraints` assigns explicit bonds and each bond adds a Morse-form energy [6] (`en_harmonic`) plus an optional harmonic/linear term; constraint-bonded pairs use the reference potential without extra Gaussians or repulsive core. Here each of the 800 water H1 must bond to one of the 200 DMSO O (1.9/2.8 Å), so `alternatenc = nint(800/200) = 4`.

Inputs

EPSR-test unchanged (`ntimes 5`, `niter 1`) except one `coordcon` line appended to the `.pcof`: `coordcon H1 1 3.0 1.0 0.2 0.5 O 1.9 2.8`. `commands.txt`: `epsr 200DMSO400Water.EPSR.inp`, `repeat 3`.

How the accuracy was verified

`commands.txt` (1 program call, repeated 3 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; only the date line of `.out` and the wall-clock columns of `fort.35` are masked). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 3 passes.
The line is rewritten by `save_files_others` as a 256-character record, so passes 2 and 3 re-read the program's own format; `coordcon.txt` (written by `fmolec1`) is compared too. Many H1 pick the same O: the O lists overflow `nbondccsize` (reallocation in pass 1), pass 2 removes the farthest H1 from over-bonded O, pass 3 re-bonds the H1 left without a bond; finally 100 % of the atoms have the expected number of bonds and `coordcon.txt` has 4-bond records. Constraint energy -1.954 kJ/mol per molecule.

Notes

Reproduced: `save_coord_energies` is only called after accepted internal rotations, so `ergold_coord` keeps its `fmolec1` value after whole-molecule moves; the pass-2 removal search stops at slot 2; stale bond-array slots are read in pass 3.

References

[1] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[3] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[4] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[5] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[6] P. M. Morse, "Diatomic molecules according to the wave mechanics. II. Vibrational levels", Phys. Rev. 34, 57-64 (1929).
[7] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.